

Axcel®
ATENOLOL
TABLET

AXCEL ATENOLOL-50MG TABLET

COMPOSITION:

Each tablet contains
Atenolol 50mg

PRESENTATION:

Off white round shaped, film coated tablet, with scoreline on one side and marked **KOTRB** on the other.

AXCEL ATENOLOL-100MG TABLET

COMPOSITION:

Each tablet contains
Atenolol 100mg

PRESENTATION:

Orange round shaped, film coated tablet, with scoreline on one side and marked with **KOTRB** on the other.

INDICATIONS:

- Control of hypertension
- Management of angina pectoris
- Control of cardiac arrhythmias
- Treatment of myocardial infarction. Early and late intervention.

PHARMACOLOGY:

Mechanism of Action:

Betablocking agents, plain selective, COTB B03. Atenolol is a beta-blocker which is beta-selective (i.e. acts preferentially on beta₁-adrenergic receptors in the heart). Selectivity decreases with increasing dose. Atenolol is without intrinsic sympathomimetic and membrane stabilising activities and as with other beta-blockers, has negative inotropic effects (and is therefore contraindicated in uncontrolled heart failure). As with other beta-blockers, the mode of action of atenolol in the treatment of hypertension is unclear. It is probably the action of atenolol in reducing cardiac rate and contractility which makes it effective in eliminating or reducing the symptoms of patients with angina. It is unlikely that any additional ancillary properties possessed by S (-) atenolol, in comparison with the racemic mixture, will give rise to different therapeutic effects. Atenolol is effective and

well-tolerated in most ethnic populations although the response may be less in black patients. Atenolol is effective for at least 24 hours after a single oral dose. The drug facilitates compliance by its acceptability to patients and simplicity of dosing. The narrow dose range and early patient response ensure that the effect of the drug in individual patients is quickly demonstrated. Atenolol is compatible with diuretics, other hypotensive agents and antianginal agents (see *Drug Interaction*). Since it acts preferentially on beta-receptors in the heart, atenolol may, with care, be used successfully in the treatment of patients with respiratory disease, who cannot tolerate non-selective beta-blockers. Early intervention with atenolol in acute myocardial infarction reduces infarct sizes and decreases morbidity and mortality. Fewer patients with a threatened infarction progress to frank infarction; the incidence of ventricular arrhythmias is decreased and marked pain relief may result in reduced need of opiate analgesics. Early mortality is decreased. Atenolol is an additional treatment to standard coronary care.

Pharmacokinetics:

Absorption of atenolol following oral dosing is consistent but incomplete (approximately 40-50%) with peak plasma concentrations occurring 2-4 hours after dosing. The atenolol blood levels are consistent and subject to little variability. There is no significant hepatic metabolism of atenolol and more than 90% of the absorbed reaches the systemic circulation unaltered. The plasma half-life is about 6 hours but this may arise in severe renal impairment since the kidney is the major route of elimination. Atenolol penetrates tissue poorly due to its low lipid solubility and its concentration in brain tissue is low. Plasma protein binding is low (approximately 3%).

DOSAGE AND ADMINISTRATION:

The dose must always be adjusted to individual requirements of the patients, with the lowest possible starting dosage.

Adults:

Hypertension

One tablet daily. Most patients respond to 100mg daily given orally as a single dose. Some patients, however, will respond to 50mg given as a single daily dose. The effect will be fully established after one to two weeks. A further reduction in blood pressure may be achieved by combining atenolol with other antihypertensive agents. For example, co-administration of atenolol with a diuretic, provides a highly effective and convenient antihypertensive therapy.

Angina

Most patients with angina pectoris will respond to 100mg given orally once or 50mg given twice daily. It is unlikely that additional benefit will be gained by increasing the dose.

Cardiac Arrhythmias

Initially controlled intravenously. Having controlled the arrhythmias with intravenous atenolol, a suitable oral maintenance dosage is 50-100mg daily, given as a single dose.

Myocardial Infarction

In suitable patients, initially controlled intravenously, followed by atenolol 50mg orally about 15 minutes later, provided no untoward effects have occurred from the intravenous dose. This should be followed by a further 50mg orally 12 hours after the intravenous dose and then 12 hours later by 100mg orally, once daily. If bradycardia and/or hypotension requiring treatment, or any other untoward effects occur, atenolol should be discontinued. Late intervention after acute myocardial infarction: For patients who present some days after suffering an acute myocardial infarction an oral dose of atenolol (100mg daily) is recommended for long-term prophylaxis of myocardial infarction.

Elderly:

Dosage requirements may be reduced, especially in patients with impaired renal function.

Children:

There is no paediatric experience with atenolol and for this reason it is not recommended for use in children.

Renal Failure:

Since atenolol is excreted via the kidneys the dosage should be reduced in cases of severe impairment of renal function. No significant accumulation of atenolol occurs in patients who have a creatinine clearance greater than 35ml/min/ 1.73m² (normal range is 100-150ml / min / 1.73m²).

For patients with a creatinine clearance of 15-35ml / min / 1.73 m² (equivalent to serum creatinine of 300-600micromol / litre) the oral dose should be 50mg daily.

For patients with a creatinine clearance of <15ml / min / 1.73m² (equivalent to serum creatinine of >600micromol / litre) the oral dose should be 25mg daily or 50mg on alternate days.

Patients on haemodialysis should be given 50mg orally after each dialysis; this should be done under hospital supervision as marked falls in blood pressure can occur.

CONTRAINDICATION:

Atenolol, as with other beta-blockers, should not be used in patients with any of the following: known hypersensitivity to the active substance, or any of the excipients; bradycardia (<45bpm); cardiogenic shock; hypotension; metabolic acidosis; severe peripheral arterial circulatory disturbances; second or third degree heart block; sick sinus syndrome; untreated phaeochromocytoma; uncontrolled heart failure.

PRECAUTION AND WARNING:

Atenolol as with other beta-blockers:

- Should not be withdrawn abruptly. The dosage should be withdrawn gradually over a period of 7-14 days, to facilitate a reduction in beta-blocker dosage. Patients should be followed during withdrawal, especially those with ischaemic heart disease.
- When a patient is scheduled for surgery, and decision is made to discontinue beta-blocker therapy, this should be done at least 24

hours prior to the procedure. The risk-benefit assessment of stopping beta-blockade should be made for each patient. If treatment is continued, an anaesthetic with little negative inotropic activity should be selected to minimise the risk of myocardial depression. The patient may be protected against vagal reactions by intravenous administration of atropine.

- Although contraindicated in uncontrolled heart failure (see *Contraindication*), may be used in patients whose signs of heart failure have been controlled. Caution must be exercised in patients whose cardiac reserve is poor.

- May increase the number and duration of angina attacks in patients with Prinzmetal's angina due to unopposed alpha-receptor mediated coronary artery vasoconstriction. Atenolol is a beta-selective beta blocker; consequently, its use may be considered although utmost caution must be exercised.

- Although contraindicated in severe peripheral arterial circulatory disturbances (see *Contraindication*), may also aggravate less severe peripheral arterial circulatory disturbances.

- Due to its negative effect on conduction time, caution must be exercised if it is given to patients with first degree heart block.

- May mask the symptoms of hypoglycaemia, in particular, tachycardia.

- May mask the signs of thyrotoxicosis.

- Will reduce heart rate, as a result of its pharmacological action. In the rare instances when a treated patient develops symptoms which may be attributable to a slow heart rate and the pulse rate drops to less than 50-55 bpm at rest, the dose should be reduced.

- May cause a more severe reaction to a variety of allergens, when given to patients with a history of anaphylactic reaction to such allergens. Such patients may be unresponsive to the usual doses of adrenaline used to treat the allergic reactions.

- May cause a hypersensitivity reaction including angioedema and urticaria.

- Should be used with caution in the elderly, starting with a lesser dose (see *Dosage and Administration*)

Since atenolol is excreted via the kidneys, dosage should be reduced in patients with a creatinine clearance of below 35ml / min / 1.7m². Although cardioselective (beta₁) beta-blockers may have less effect on lung function than non-selective beta-blockers, as with all, beta-blockers, these should be avoided in patients with reversible obstructive airways disease, unless there are compelling clinical reasons for their use. Where such reasons exist, atenolol may be used with caution. Occasionally, some increase in airways resistance may occur in asthmatic patients, however, and this may usually be reversed by commonly used dosage of bronchodilators such as salbutamol or isoprenaline. As with other beta-blockers, in patients with a phaeochromocytoma, an alpha-blocker should be given concomitantly.

Use in Pregnancy and Lactation:

Atenolol crosses the placental barrier and appears in the cord blood. No studies have been performed on the use of atenolol in the first trimester and the possibility of foetal injury cannot be

excluded. Atenolol has been used under close supervision for the treatment of hypertension in the third trimester. Administration of atenolol to pregnant women in the management of mild to moderate hypertension has been associated with intra-uterine growth retardation. The use of atenolol in women who are, or may become, pregnant requires that the anticipated benefit be weighed against the possible risks, particularly in the first and second trimesters, since beta-blockers, in general, have been associated with a decrease in placental perfusion which may result in intra-uterine deaths, immature and premature deliveries. There is significant accumulation of atenolol in breast milk. Neonates born to mothers who are receiving atenolol at parturition or breast-feeding may be at risk for hypoglycaemia and bradycardia. Caution should be exercised when atenolol is administered during pregnancy or to a woman who is breast-feeding.

Effect on ability to drive and use machines:
Use is unlikely to result in any impairment of the ability of patients to drive or operate machinery. However it should be taken into account that occasionally dizziness or fatigue may occur.

SIDE EFFECTS:
Atenolol is well tolerated. In clinical studies, the undesired events reported are usually attributable to the pharmacological actions of atenolol. The following undesired events, listed by body system, have been reported with the following frequencies: Very Common (≥10%), common (1-9.9%), uncommon (0.1-0.9%), rare (0.01-0.09%), very rare (<0.01%) including isolated reports.

Cardiac disorders:
Common: Bradycardia.
Rare: Heart failure deterioration, precipitation of heart block.

Vascular disorders:
Common: Cold extremities.
Rare: Postural hypotension which may be associated with syncope, intermittent claudication may be increased if already present, in susceptible patients Raynaud's phenomenon.

Nervous system disorders:
Rare: Dizziness, headache, paraesthesia.

Psychiatric disorders:
Uncommon: Sleep disturbances of the type noted with other beta blockers.
Rare: Mood changes, nightmares, confusion, psychoses and hallucinations.

Gastrointestinal disorders:
Common: Gastrointestinal disturbances.
Rare: Dry mouth.

Investigations:
Uncommon: Elevations of transaminase levels.
Very rare: An increase in ANA (Antinuclear Antibodies) has been observed, however the clinical relevance of this is not clear.

Hepatobiliary disorders:
Rare: Hepatic toxicity including intrahepatic cholestasis.

Blood and lymphatic system disorders:
Rare: Purpura, thrombocytopenia.

Skin and subcutaneous tissue disorder:
Rare: Alopecia, psoriasisform skin reactions, exacerbation of psoriasis, skin rashes.
Not known: Hypersensitivity reactions, including angioedema and urticaria.

Eye disorders:
Rare: Dry eyes, visual disturbances.

Reproductive system and breast disorders:
Rare: Impotence.

Respiratory, thoracic and mediastinal disorders:
Rare: Bronchospasm may occur in patients with bronchial asthma or a history of asthmatic complaints.

General disorders and administration site conditions:
Common: Fatigue.

Discontinuance of the drug should be considered if, according to clinical judgement, the well- being of the patients is adversely affected by any of the above reactions.

DRUG INTERACTION:
Combined use of beta-blockers and calcium channel blockers with negative inotropic effects e.g. verapamil, diltiazem can lead to an exaggeration of these effects particularly in patients with impaired ventricular function and/or sino-atrial or atrio-ventricular conduction abnormalities. This may result in severe hypotension, bradycardia and cardiac failure. Neither the beta-blocker nor the calcium channel blocker should be administered intravenously within 48 hours of discontinuing the other. Concomitant therapy with dihydropyridines e.g. nifedipine, may increase the risk of hypotension, and cardiac failure may occur in patients with latent cardiac insufficiency. Digitalis glycosides, in association with beta-blockers, may increase atrio-ventricular conduction time. Beta-blockers may exacerbate the rebound hypertension, which can follow the withdrawal of clonidine. If the two drugs are co-administered, the beta-blocker should be withdrawn several days before discontinuing clonidine. If replacing clonidine by beta-blocker therapy, the introduction of beta-blockers should be delayed for several days after clonidine administration has stopped.

Class I anti-arrhythmic drugs (e.g. disopyramide) and amiodarone may have a potentiating effect on atrial-conduction time and induce negative inotropic effect. Concomitant use of sympathomimetic agents e.g. adrenaline, may counteract the effect of beta-blockers. Concomitant use with insulin and oral anti-diabetic drugs may lead to the intensification of the blood sugar lowering effects of these drugs. Symptoms of hypoglycaemia, particularly tachycardia, may be masked (see *Precaution and Warning*). Concomitant use of prostaglandin synthetase inhibiting drugs (e.g. ibuprofen, indomethacin) may decrease the hypotensive effects of beta-blockers. Caution must be exercised when using anaesthetic agents with atenolol. The anaesthetist should be informed and the choice of anaesthetic should be an agent with as little negative inotropic activity as possible. Use of beta-blockers with anaesthetic drugs may result in attenuation of the reflex tachycardia and increase the risk of hypotension. Anaesthetic agents causing myocardial depression are best avoided.

OVERDOSAGE AND TREATMENT:
The symptoms of overdosage may include bradycardia, hypotension, acute cardiac insufficiency and bronchospasm. General treatment should include: close supervision, treatment in an intensive care ward, the use of gastric lavage, activated charcoal and a laxative to prevent absorption of any drug still present in the gastrointestinal tract, the use of plasma or plasma substitutes to treat hypotension and shock. The use of haemodialysis or haemoperfusion may be considered. Excessive bradycardia can be countered with atropine 1-2mg intravenously and/or cardiac pacemaker. If necessary, this may be followed by a bolus dose of glucagon 10mg intravenously. If required, this may be repeated or followed by an intravenous infusion of glucagon 1-10mg / hour depending on the response. If no response to glucagon occurs or if glucagon is unavailable, a beta-adrenoreceptor stimulant such as dobutamine 2.5 to 10 micrograms / kg / minute by intravenous infusion may be given. Dobutamine, because of its positive inotropic effect could also be used to treat hypotension and acute cardiac insufficiency. It is likely that these doses would be inadequate to reverse the cardiac effects of beta-blocker blockade if a large overdose has been taken. The dose of dobutamine should therefore be increased if necessary to achieve the required response according to the clinical condition of the patient. Bronchospasm can usually be reversed by bronchodilators.

STORAGE:
Do not store above 30°C. Protect from light and moisture.

KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACK QUANTITIES:
Available in blister pack of 28's.

Further information can be obtained from pharmacist, physician or the manufacturer.



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